Advanced Data Structures & Algorithm Analysis (2501CS10)

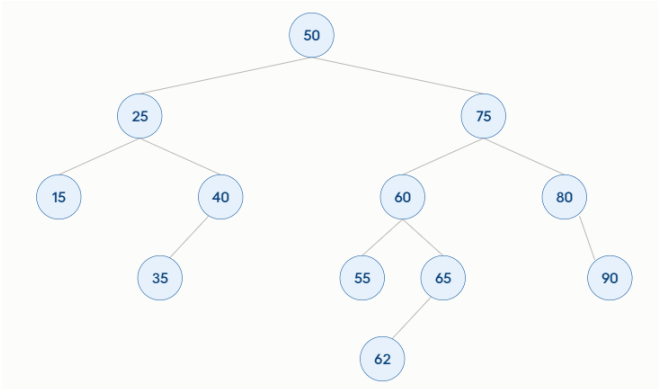
Unit-wise Important Questions

UNIT-I:

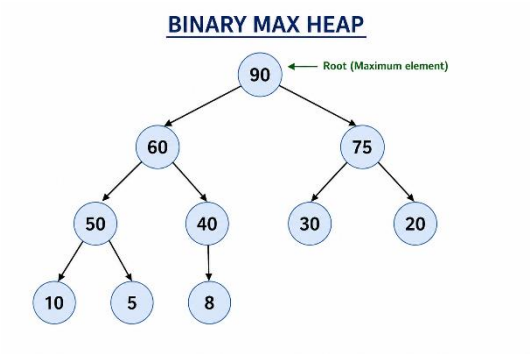
1. Explain the importance of algorithm analysis. Differentiate between time complexity and space complexity with suitable examples.
2. Describe the asymptotic notations **O, Ω, and Θ**. Illustrate each notation with suitable examples.
3. Analyze the time complexity for matrix multiplication algorithm (tabular method) using asymptotic notation and justify your answer.
4. Explain the concept of hashing. Discuss different hash functions with suitable examples.
5. Explain the collision resolution techniques Linear Probing, Quadratic Probing, Double Hashing, and Separate Chaining.
6. Differentiate between Static Hashing and Dynamic Hashing. Explain the working of Dynamic Hashing (Extendable hashing) with an example.

UNIT-II:

1. Construct an AVL Tree by inserting the following keys: 50, 25, 12, 80, 90, 37, 45, 18, 19, 6. Show the required rotations.
2. Explain the four types of AVL Tree rotations with suitable examples. Discuss the applications of AVL Trees.
3. Explain the deletion procedure in AVL tree. Show the balancing operations after deletion of 15 from the below AVL tree.



4. Construct a B-Tree of order 4 by inserting the following keys: 5, 3, 21, 9, 1, 13, 2, 7, 10, 12, 4, 8.
5. Explain the deletion operation in a B-Tree with a suitable example, and applications of B-Trees.
6. Construct a Min/Max Heap by inserting the following elements: 40, 15, 30, 10, 12, 6, 25, 18. Show all intermediate steps.
7. Perform one deletion operation on given Max Heap and show the step-by-step process for final Max Heap.



8. Explain the basic operations of a Binary Heap with their time complexities and also discuss the applications of Priority Queues.